

## Supplementary material

**Table S1.** Detailed search strategies for each electronic database

| Data base  | Research strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Date     | Results |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|
| Medline #1 | ("Methylene Blue"[Mesh]<br>OR "methylene blue"[tiab]<br>OR "methylthioninium chloride"[tiab])<br>AND<br>("Shock"[Mesh]<br>OR shock[tiab]<br>OR "circulatory shock"[tiab]<br>OR "septic shock"[tiab]<br>OR "cardiogenic shock"[tiab]<br>OR "vasodilatory shock"[tiab]<br>OR "vasoplegic shock"[tiab]<br>OR vasoplegia[tiab]<br>OR "hypovolemic shock"[tiab]<br>OR "distributive shock"[tiab]<br>OR "obstructive shock"[tiab]<br>OR "anaphylactic shock"[tiab]<br>OR "neurogenic shock"[tiab]<br>OR "refractory shock"[tiab]<br>OR "postoperative shock"[tiab])                                                                                             | 01/02/26 | 515     |
| Medline #2 | ((((((Methylene blue[Title/Abstract]) AND<br>(shock[Title/Abstract])) OR ( septic<br>shock[Title/Abstract])) OR ( vasoplegia<br>syndrome[Title/Abstract])) NOT (children)) NOT<br>(neonates) AND (2007:2026[pdat])) NOT (case<br>report)                                                                                                                                                                                                                                                                                                                                                                                                                  | 01/02/26 | 189     |
| Embase     | #20 #3 AND #19 912<br>#19 #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR<br>#10 OR #11 OR #12 OR #13 OR #14 OR #15 OR<br>#16 OR #17 OR #18 330450<br>#18 'postoperative shock':ab,ti,kw 178<br>#17 'refractory shock':ab,ti,kw 1561<br>#16 'neurogenic shock':ab,ti,kw 281<br>#15 'anaphylactic shock':ab,ti,kw 5925<br>#14 'obstructive shock':ab,ti,kw 612<br>#13 'distributive shock':ab,ti,kw 780<br>#12 'hypovolemic shock':ab,ti,kw 4741<br>#11 'vasoplegia' 1918<br>#10 'vasoplegia':ab,ti,kw 1239<br>#9 'vasoplegic shock':ab,ti,kw 356<br>#8 'vasodilatory shock':ab,ti,kw 693<br>#7 'cardiogenic shock':ab,ti,kw 37161<br>#6 'septic shock':ab,ti,kw 56313 | 03/02/26 | 912     |

|                  |    |                                                                                                                                                                                                                                                                                                                                                               |          |    |
|------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----|
|                  | #5 | 'circulatory shock':ab,ti,kw                                                                                                                                                                                                                                                                                                                                  | 2008     |    |
|                  | #4 | 'shock':ab,ti,kw                                                                                                                                                                                                                                                                                                                                              | 329321   |    |
|                  | #3 | #1 OR #2                                                                                                                                                                                                                                                                                                                                                      | 30701    |    |
|                  | #2 | 'methylthioninium chloride':ab,ti,kw                                                                                                                                                                                                                                                                                                                          | 119      |    |
|                  | #1 | 'methylene blue':ab,ti,kw                                                                                                                                                                                                                                                                                                                                     | 30633    |    |
|                  |    | ("methylene blue" OR methylthioninium OR<br>methylthioninium chloride OR methylen*<br>blue*)<br>AND<br>(shock* OR septic shock* OR septic* OR<br>distributive shock* OR vasodilatory shock*<br>OR cardiogenic shock* OR hypovolemic shock*<br>OR hemorrhagic shock*<br>OR anaphylactic shock* OR anaphylax* OR<br>neurogenic shock*<br>OR obstructive shock*) |          |    |
| Cochrane Central |    |                                                                                                                                                                                                                                                                                                                                                               | 01/02/26 | 88 |

**Table S2.** Full text review excluded articles and reasons for exclusion.

| Reference                                                                       | Reasons for exclusion                         |
|---------------------------------------------------------------------------------|-----------------------------------------------|
| Juffermans et al. Nitric Oxide 22 (2010) 275–280(1)                             | Wrong study design                            |
| Mehaffey et al. Ann Thorac Surg 2017;104:36–41(2)                               | Wrong study design                            |
| Donati et al. Crit Care Med 2002 Vol. 30, No. 10 (3)                            | Wrong study design                            |
| Hiruy et al. J of Cardiothor and Vasc Anesth 37<br>(2023) 2228 2235(4)          | Wrong study design                            |
| Gachot et al. Intensive Care Med (1995) 21:1027-<br>1031(5)                     | Wrong study design                            |
| Yune, H et al. 1453: Crit Care Med 44(12):p 439,<br>2016.(6)                    | Wrong publication type                        |
| Delgadillo NA. Crit Care Med. 2023;51(1):22-22. (7)                             | Wrong publication type                        |
| Ponnappan KT, et al. InTransplantation 2018 May<br>1 (Vol. 102, pp. 38-39). (8) | Wrong publication type (No results published) |

<sup>1</sup> Juffermans NP, Vervloet MG, Daemen-Gubbels CRG, Binnekade JM, Jong MD, Groeneveld ABJ. A dose-finding study of methylene blue to inhibit nitric oxide actions in the hemodynamics of human septic shock. Nitric Oxide. mayo de 2010;22(4):275-80. doi:10.1016/j.niox.2010.01.006

2. Mehaffey JH, Johnston LE, Hawkins RB, Charles EJ, Yarboro L, Kern JA, et al. Methylene Blue for Vasoplegic Syndrome After Cardiac Operation: Early Administration Improves Survival. *Ann Thorac Surg.* julio de 2017;104(1):36-41. doi:10.1016/j.athoracsur.2017.02.057
3. Donati A, Conti G, Loggi S, Münch C, Coltrinari R, Pelaia P, et al. Does methylene blue administration to septic shock patients affect vascular permeability and blood volume?: *Crit Care Med.* octubre de 2002;30(10):2271-7. doi:10.1097/00003246-200210000-00015
4. Hiruy A, Ciapala S, Donaldson C, Wang L, Hohlfelder B. Hydroxocobalamin Versus Methylene Blue for the Treatment of Vasoplegic Shock Associated With Cardiopulmonary Bypass. *J Cardiothorac Vasc Anesth.* noviembre de 2023;37(11):2228-35. doi:10.1053/j.jvca.2023.07.015
5. Gachot B, Bédos JP, Veber B, Wolff M, Régnier B. Short-term effects of methylene blue on hemodynamics and gas exchange in humans with septic shock. *Intensive Care Med.* diciembre de 1995;21(12):1027-31. doi:10.1007/BF01700666
6. Yune H, Kim K, Jo YH, Kim J, Lee JH, Chung H, et al. 1453: INFUSION OF METHYLENE BLUE IN SEVERE SEPSIS AND SEPTIC SHOCK: A RANDOMIZED CONTROLLED TRIAL. *Crit Care Med.* diciembre de 2016;44(12):439-439. doi:10.1097/01.ccm.0000510127.34774.13
7. Delgadillo NA, Ibarra-Estrada M, Sandoval-Plascencia L, López-Pulgarin JA, Chavez-Peña Q, Mijangos-Mendez J, et al. 44: ADJUNCTIVE METHYLENE BLUE FOR PATIENTS IN SEPTIC SHOCK: THE SHOCKEM-BLUE TRIAL. *Crit Care Med.* enero de 2023;51(1):22-22. doi:10.1097/01.ccm.0000906052.35064.ea
8. KT Ponnappan, Pandey CK, Maiwall R, Saluja V, Tandon M, Parvez MQ, et al. Evaluating in cirrhotics with refractory vasoplegia the effect of methylene blue (crumbs)-a randomized controlled trial. *InTransplantation.* mayo de 2018;102:38-9.